Fundamentals of Optics

Lecturer

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Room 239, South Tower, College of Engineering

Rules

- **Final exam: 30%**
- **Mid-term exam: 30%**
- **Homework: 25%**
- **Attendance: 15%**

Reference books

Optics, by Eugene Hecht

光学，赵凯华、钟锡华

群名称：2025春季光学基础

群　号：1033369719

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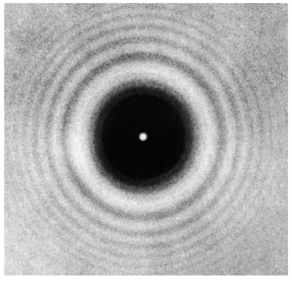
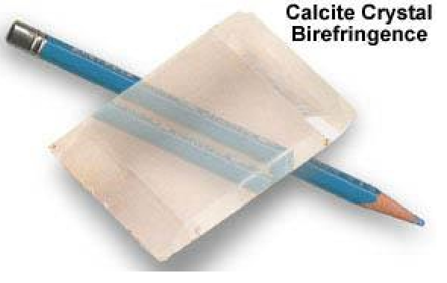
A brief history

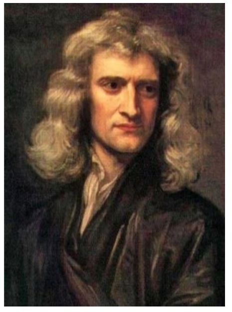
- ❖ The nature of light
- ❖ In the seventeenth century
- ✓ It travelled in straight lines;
- ✓ It was reflected off smooth surfaces;
- ✓ It changed direction when it passed from one medium to another (established by Snell);
- ✓ What we now call Fresnel diffraction had been discovered;
- ✓ Double refraction had been discovered.


Diffraction



Calcite crystal birefringence

The wave-corpuscle controversy

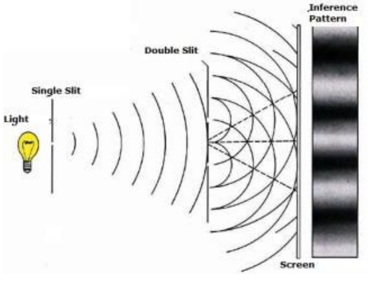


Newton
Corpuscular theory

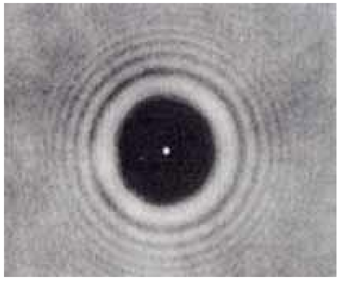
Huygens
Wave theory

Triumph of the wave theory

- In 1802 Young demonstrated interference fringes between waves from two sources;
- In 1821 Fraunhofer invented the diffraction grating and produced diffraction patterns in parallel light;
- In 1818 Fresnel and Arago found that the bright spot at the centre of the shadow of a disc.



Young's interference experiment



Poisson's spot 泊松亮斑

The nature of light waves: Transverse or longitudinal?

- The greatest step towards understanding the waves came from the theoretical study of magnetism and electricity.
  - ✓ In the first half of the nineteenth century the relationship between magnetism and electricity had been worked out fairly thoroughly, by men such as Oersted, Ampere and Faraday.
  - ✓ In 1865, Maxwell was inspired to combine his predecessors' observations in mathematical form by describing the region of influence around electric charges and magnets as an "electromagnetic field" and expressing the observations in terms of differential equations. In manipulating these equations he found that they could assume the form of **a transverse wave equation**.
- Light was established as an electromagnetic field.

Quantum theory

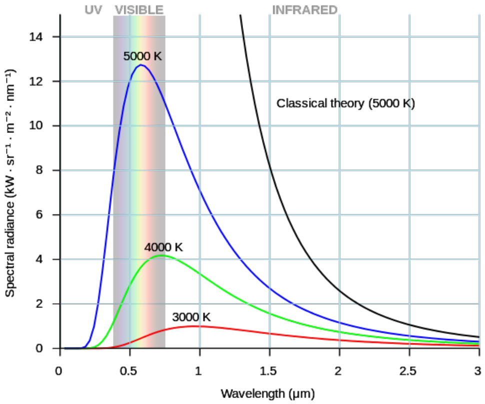
- There remained some basic problems, as the study of the light emitted by hot bodies indicated.

$$M_{0(\lambda T)} = \frac{2\pi CKT}{\lambda^4}$$

Rayleigh-Jean's Formula isn't fit the experimental points at short wavelength.

$$M_{0(\lambda T)} = \frac{c_1}{\lambda^5} e^{-\frac{c_2}{\lambda T}}$$

Wien's formula isn't fit to the experimental points at long wavelength.



Black-body radiation

In 1900 Max Planck pointed out that if Wien's formula were modified in a simple way it would prove to fit the data precisely. It is:

$$M_{0(\lambda T)} = 2\pi hC^2 \lambda^{-5} \frac{1}{e^{\frac{hC}{k\lambda T}} - 1}$$

$$= 2\pi hC^2 \lambda^{-5} \frac{1}{e^{\frac{h\nu}{kT}} - 1}$$

- Planck's idea: the wave energy is divided into packets, now called photons, whose energy content is proportional to the frequency.
- By about 1920 it was generally accepted, largely based on:
  - Einstein's study on the photo-electric effect.
  - Compton's understanding of energy and momentum conservation in the scattering of X-rays by electrons.

Quantum theory (cont.)

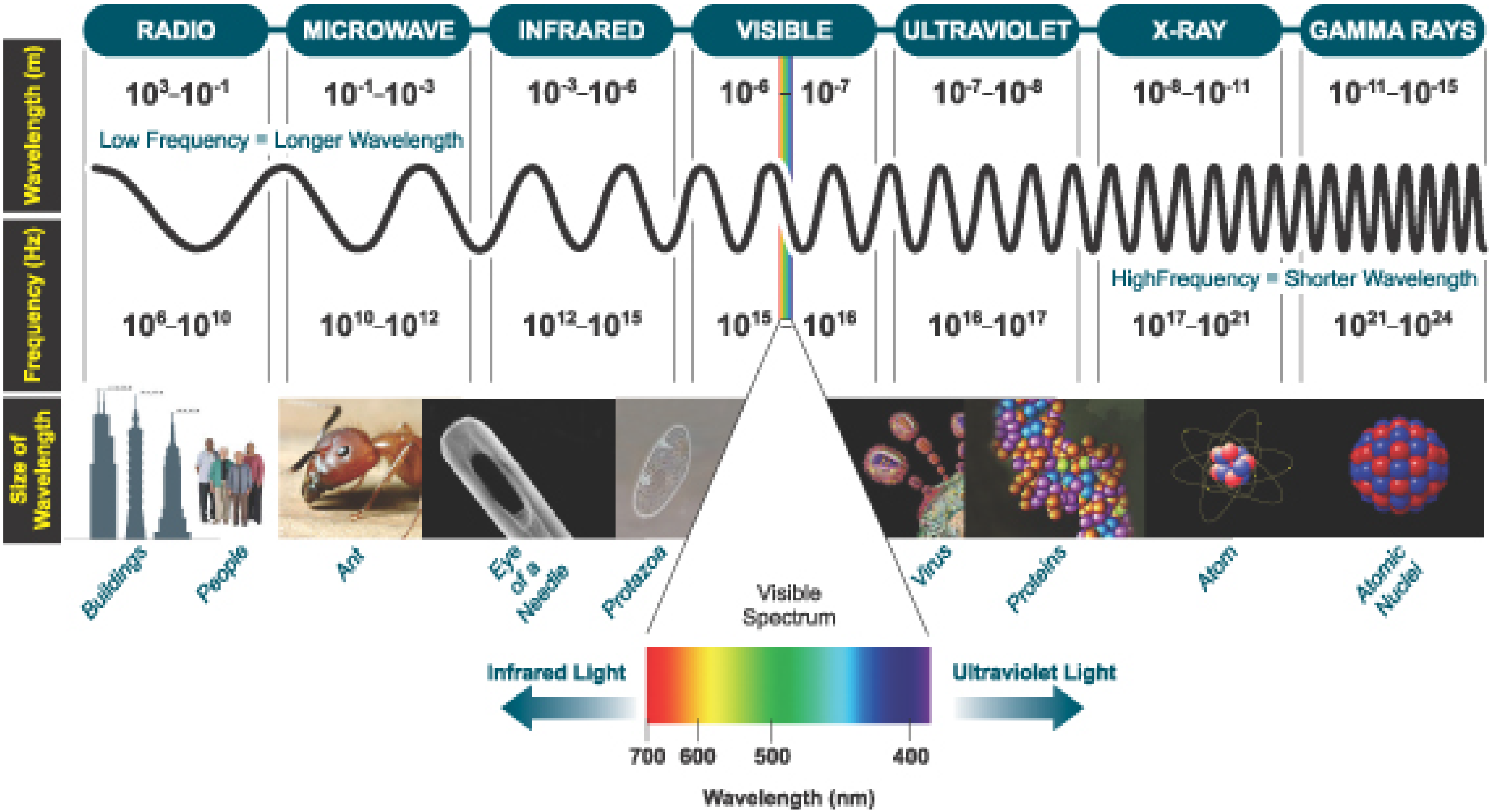
- **Wave-particle duality**
  - A way of understanding duality questions in linear optics
    - The wave intensity tells us the probability of finding a photon at any given point.
    - The corpuscular features only arise when the wave interacts with a medium and give up its energy to it.
- **Corpuscular waves**
  - Faraday "if electricity produces magnetism, does magnetism produce electricity?"
  - De Broglie "if waves are corpuscles, are corpuscles waves?"
    - Davisson and Germer by ionization methods and G.P. Thomson by photographic methods showed that fast-moving electrons could be diffracted by matter similarly to X-rays.
    - Schrodinger in 1928 produced a general wave theory of matter.

About this course

- The physics of waves
- Electromagnetic theory, photons and light
- The propagation of light
- Geometrical optics
- The superposition of waves (including Fourier optics)
- Polarization
- Interference (including coherence)
- Diffraction

CHAPTER I
WAVE MOTION

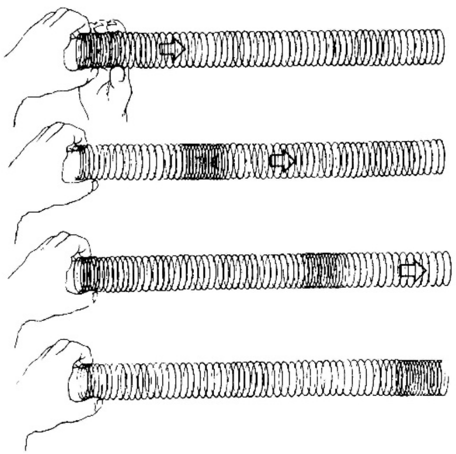
Waves



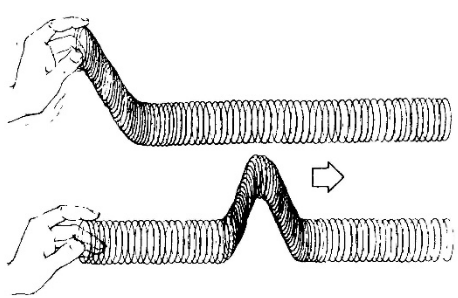
RADIO
MICROWAVE
INFRARED
VISIBLE
ULTRAVIOLET
X-RAY
GAMMA RAYS
Wavelength (m)
$10^{3}-10^{-1}$
$10^{-1}-10^{-3}$
$10^{-3}-10^{-6}$
$10^{-6} - 10^{-7}$
$10^{-7}-10^{-8}$
$10^{-8}-10^{-11}$
$10^{-11}-10^{-15}$
Low Frequency = Longer Wavelength
HighFrequency = Shorter Wavelength
Frequency (Hz)
$10^{6}-10^{10}$
$10^{10}-10^{12}$
$10^{12}-10^{15}$
$10^{15} - 10^{16}$
$10^{16}-10^{17}$
$10^{17}-10^{21}$
$10^{21}-10^{24}$
Size of Wavelength
Buildings
People
Ant
Eye of a Needle
Protozoa
Virus
Proteins
Atom
Atomic Nuclei
Visible Spectrum
Infrared Light
Ultraviolet Light
700
600
500
400
Wavelength (nm)

One-dimensional wave

An essential aspect of a traveling wave is that it is a self-sustaining disturbance of the medium through which it propagates.



(a)

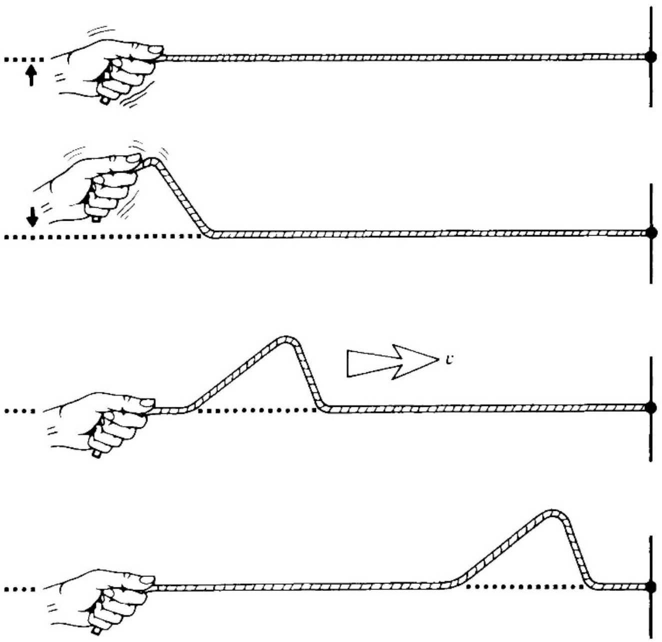


(b)

(a) A longitudinal wave in a spring.
(b) A transverse wave in a spring.

Longitudinal wave: the medium is displaced in the direction of motion of the wave, such as sound waves.
Transverse wave: the medium is displaced in a direction perpendicular to that of the motion of the wave, such as electromagnetic waves.

❖ *The disturbance advances, NOT the material medium.*



A wave on a string

- **Envision some such disturbance**

ψ moving in the position x-direction with a constant speed v.

- **Disturbance function:**

$$\psi(x,t) = f(x,t)$$

where $f(x,t)$ corresponds to some specific function or wave shape.

- **For t=0**

$$\psi(x,t)|_{t=0} = f(x,0) = f(x)$$

It represents the profile of the wave at that time.

- Introduce a coordinate system S', that travels along with the pulse at the speed υ.

$$\psi = f(x')$$

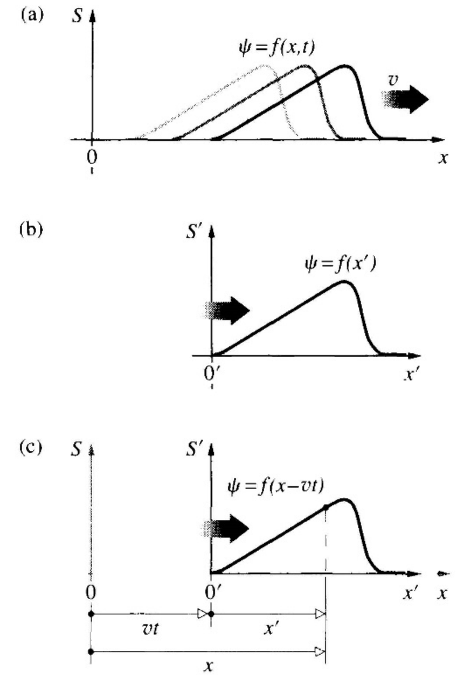
- Due to

$$x' = x - \upsilon t$$

- Then we have

$$\psi(x,t) = f(x - \upsilon t)$$

- This represents the most general form of the one-dimensional **wavefunction**, which moves in the positive *x*-direction with a speed υ.



Moving reference frame

- Exam ψ after an increase in time of Δt and a corresponding increase of $\upsilon\Delta t$ in *x*:

$$f[(x+\upsilon\Delta t)-\upsilon(t+\Delta t)] = f(x-\upsilon t)$$

The profile is unaltered.

- The wave traveling in the negative *x*-direction

$$\psi(x,t) = f(x+\upsilon t)\ with\ \upsilon>0$$

1.1 The differential wave equation

- Take the partial derivative of $\psi(x,t) = f(x')$ with respect to *x*, holding *t* constant.
- Using $x' = x - vt$

$$\frac{\partial \psi}{\partial x} = \frac{\partial f}{\partial x}$$

$$\frac{\partial \psi}{\partial x} = \frac{\partial f}{\partial x'}\frac{\partial x'}{\partial x} = \frac{\partial f}{\partial x'} \quad \text{.......(a)}$$

Because $\quad \frac{\partial x'}{\partial x} = \frac{\partial(x \mp vt)}{\partial x} = 1$

- Holding *x* constant, the partial derivative with respect to time

$$\frac{\partial \psi}{\partial t} = \frac{\partial f}{\partial x'}\frac{\partial x'}{\partial t} = \frac{\partial f}{\partial x'}(\mp v) = \mp v\frac{\partial f}{\partial x'} \quad \text{.........(b)}$$

- Combining (a) and (b), we have

$$\frac{\partial \psi}{\partial t} = \mp v\frac{\partial \psi}{\partial x}$$

- The second partial derivatives:

$$\frac{\partial^2\psi}{\partial x^2} = \frac{\partial^2 f}{\partial x'^2} \qquad \text{.........(c)}$$

- $$\frac{\partial^2\psi}{\partial t^2} = \frac{\partial}{\partial t}\left(\mp v\frac{\partial f}{\partial x'}\right) = \mp v\frac{\partial}{\partial x'}\left(\frac{\partial f}{\partial t}\right) = v^2\frac{\partial^2 f}{\partial x'^2} \qquad \text{.........(d)}$$

- Combining (c) and (d), we have

$$\boldsymbol{\frac{\partial^2\psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}}$$

- Homogeneous differential equation (齐次微分方程)
- Undamped systems (无阻尼系统)

which is the desired one-dimensional **differential wave equation**.

1.2 Harmonic wave (谐波)

- The simplest wave form: a sine or cosine curve, known as harmonic waves.

  $\psi(x,t)|_{t=0} = \psi(x) = Asin(kx) = f(x)$

  - k is a positive constant known as the propagation number.
  - A is the maximum disturbance known as amplitude of the wave.

- A progressive wave traveling at speed v in the positive x-direction [just replace x with $(x-vt)$]:

  $\psi(x,t) = Asink(x-vt) = f(x-vt)$

Spatial period

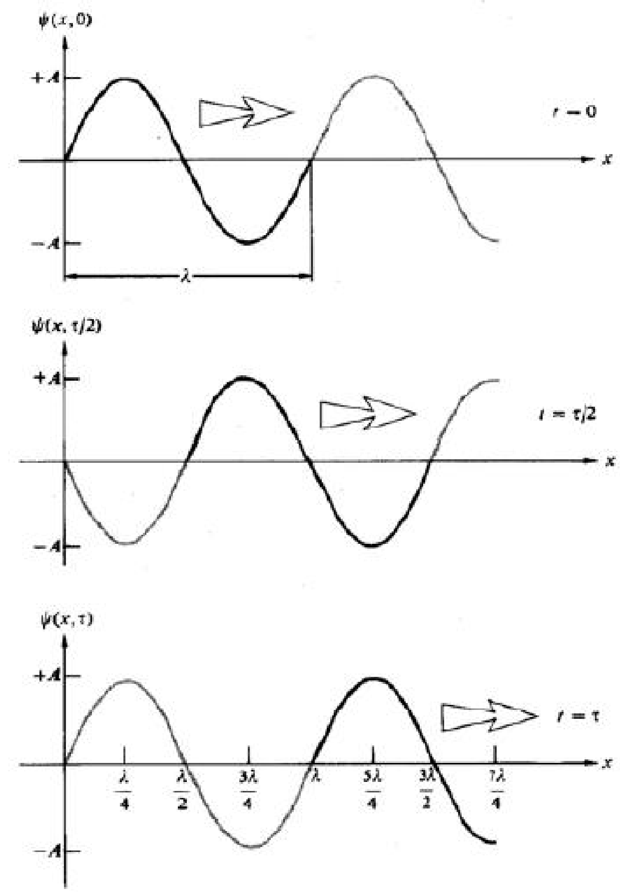
- The spatial period is known as the wavelength, λ. **Wavelength** is *__the number of units of length per wave__*.
- An increase or decrease in *x* by the amount λ should leave ψ unaltered:

  $$\psi(x,t)=\psi(x\pm\lambda,t)$$

- For a harmonic wave:

  $$\sin k(x-vt) = \sin k[(x\pm\lambda)-vt]$$
  $$= \sin [k(x-vt)\pm 2\pi]$$

- So $k=2\pi/\lambda$



Temporal period

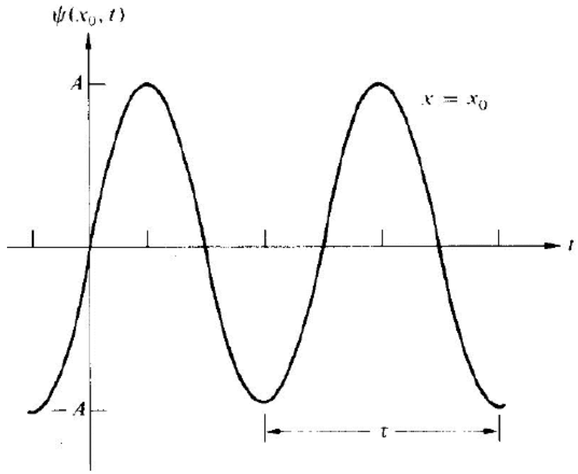
- Temporal period (τ): the amount of time it takes for one complete wave to pass a stationary observer.

$$\psi(x,t) = \psi(x,t\pm\tau)$$

- For a harmonic wave

$$\sin k(x-\upsilon t) = \sin k[x-\upsilon(t\pm\tau)]$$
$$= \sin [k(x-\upsilon t)\pm 2\pi]$$

- We have $\tau=\lambda/\upsilon$



Spatial and temporal period (cont.)

- Temporal frequency ν: the number of waves per unit of time (unit: cycles/s or Hertz)

  $$\nu \equiv 1/\tau$$

- Angular temporal frequency (unit: radians/s)

  $$\omega \equiv 2\pi/\tau = 2\pi\nu$$

- Wave number: the number of waves per unit of length (m^{-1})

  $$\kappa \equiv 1/\lambda$$

  Example: the energy of a photon

  $$E = h\nu = \hbar\omega$$

  $$E = h\nu = hc/\lambda = hc\kappa$$

  where *h* is Planck's constant.

- Another kind of equivalent expression:

$$\psi(x,t) = A\sin(kx \mp \omega t)$$

Different forms of equivalent expression

- $\psi(x,t) = Asink(x \mp \upsilon t)$
- $\psi(x,t) = Asin2\pi(\frac{x}{\lambda} \mp \frac{t}{\tau})$
- $\psi(x,t) = Asin2\pi(\kappa x \mp \nu t)$
- $\psi(x,t) = Asin(kx \mp \omega t)$
- $\psi(x,t) = Asin\ 2\pi\nu(\frac{x}{\upsilon} \mp t)$

1.3 Phase and phase velocity

- General harmonic wavefunction:
  $\psi(x,t) = A sin(kx - \omega t + \varepsilon)$
  where ε is the initial phase
- The phase of a disturbance:
  $\varphi(x,t) = kx - \omega t + \varepsilon$
- The speed of propagation on the condition of constant phase:

$$\left(\frac{\partial x}{\partial t}\right)_{\varphi} = \frac{-\left(\frac{\partial \varphi}{\partial t}\right)_{x}}{\left(\frac{\partial \varphi}{\partial x}\right)_{t}} = \pm\frac{\omega}{k} = \pm\upsilon$$

- This is the speed at which the profile moves and is known as the ***phase velocity*** of the wave.

1.4 The superposition principle

- Suppose that the wavefunctions $\psi 1$ and $\psi 2$ are each separate solutions of the wave equation, it follows that ($\psi 1+\psi 2$) is also the solution.

$$\frac{\partial^2\psi_1}{\partial x^2}=\frac{1}{\upsilon^2}\frac{\partial^2\psi_1}{\partial t^2} \quad \text{and} \quad \frac{\partial^2\psi_2}{\partial x^2}=\frac{1}{\upsilon^2}\frac{\partial^2\psi_2}{\partial t^2}$$

So $$\frac{\partial^2\psi_1}{\partial x^2}+\frac{\partial^2\psi_2}{\partial x^2}=\frac{1}{\upsilon^2}\frac{\partial^2\psi_1}{\partial t^2}+\frac{1}{\upsilon^2}\frac{\partial^2\psi_2}{\partial t^2}$$

We have $$\frac{\partial^2}{\partial x^2}(\psi_1+\psi_2)=\frac{1}{\upsilon^2}\frac{\partial^2}{\partial t^2}(\psi_1+\psi_2)$$

- The resulting disturbance at each point in the region of overlap is the algebraic sum of the individual constituent waves at that location.

1.5 Complex representation

- The complex number $\tilde{z}$

$$\tilde{z} = x + iy$$

- In terms of polar coordinates (r, θ)

$$\tilde{z} = x + iy = r(\cos\theta + i\sin\theta)$$

- According to the Euler formula

$$\tilde{z} = re^{i\theta} = r(\cos\theta + i\sin\theta)$$

- The wavefunction is written as

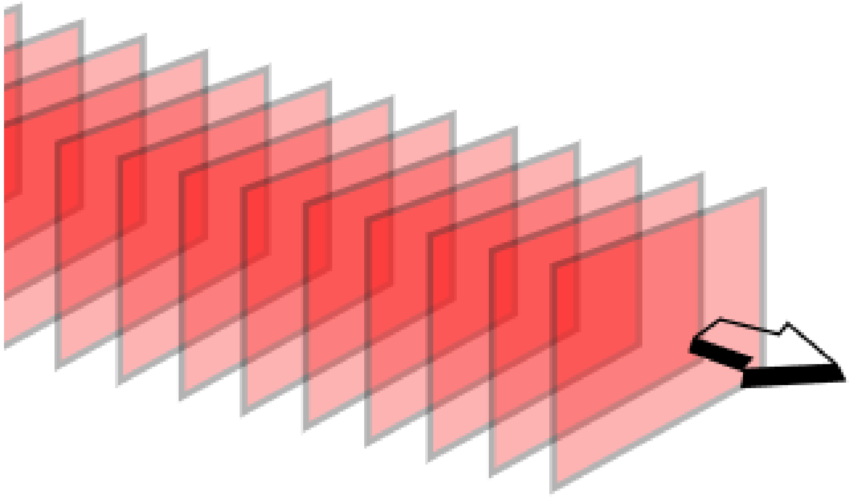
$$\psi(x,t) = Ae^{i(kx-\omega t+\varepsilon)} = Ae^{i\varphi}$$

- It is customary to choose the real part to describe a harmonic wave

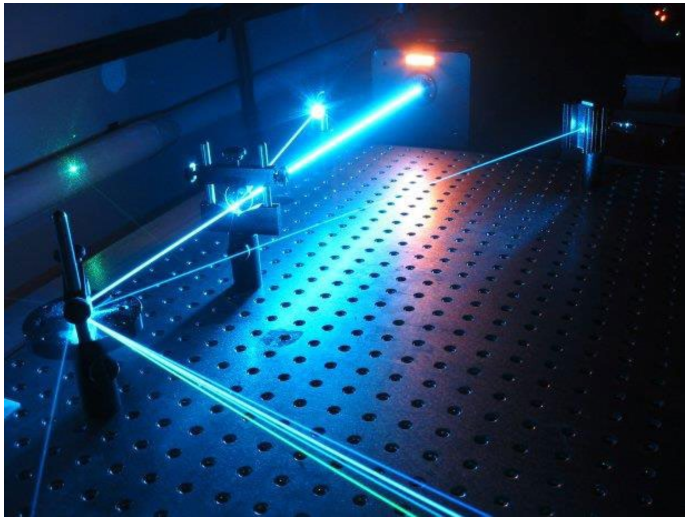
$$\psi(x,t) = Re\left[Ae^{i(kx-\omega t+\varepsilon)}\right] = A\cos(kx-\omega t+\varepsilon)$$

1.6 Plane waves

The plane waves: at a given time, all the surfaces on which a disturbance has constant phase form a set of planes, each perpendicular to the propagation direction.



The wavefronts of a plane wave traveling in 3D space



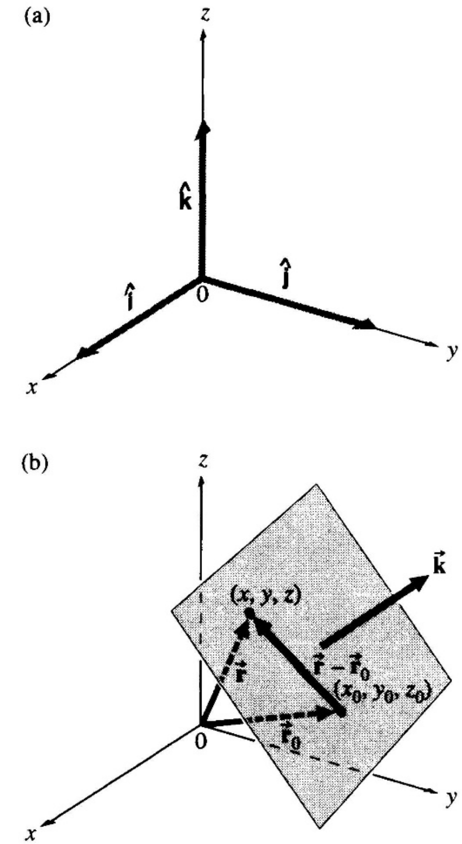
Laser beams

- In Cartesian coordinates:

  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

  $(\vec{r} - \vec{r}_0) = (x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}$
- By setting: $(\vec{r} - \vec{r}_0) \cdot \vec{k} = 0$
- So we force the vector $(\vec{r} - \vec{r}_0)$ to sweep out a plane perpendicular to $\vec{k}$.
- With $\vec{k} = k_x\hat{i} + k_y\hat{j} + k_z\hat{k}$
- So $k_x(x - x_0) + k_y(y - y_0) + k_z(z - z_0) = 0$
- $k_x x + k_y y + k_z z = a$

  where $a = k_x x_0 + k_y y_0 + k_z z_0 = constant$
- So $\vec{k} \cdot \vec{r} = constant = a$
- The plane is the locus of all points whose position vectors each have the same projection onto the $\vec{k}$-direction



- Construct a set of planes over which $\psi(r)$ varies in space sinusoidally:

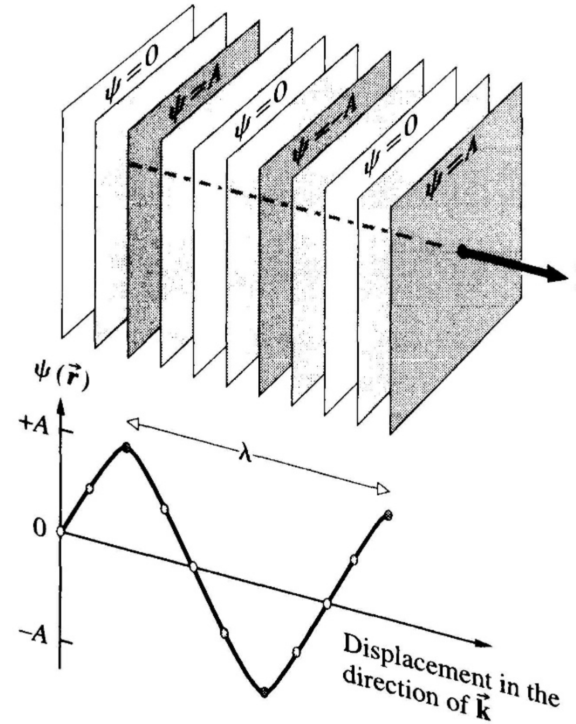
$$\psi(\vec{r}) = Ae^{i\vec{k}\cdot\vec{r}}$$

where k=$2\pi/\lambda$

- The vector k is called the propagation vector.
- As to the traveling wave:

$$\psi(\vec{r}) = Ae^{i\left(\vec{k}\cdot\vec{r}\mp\omega t\right)}$$

- This disturbance travels along in the $\vec{k}$-direction. At any given time, the surfaces joining all points of equal phase are known as wavefronts.



- The phase velocity of a plane wave:

$$\psi(\vec{r},t) = \psi(r_k + dr_k, t + dt) = \psi(r_k, t)$$

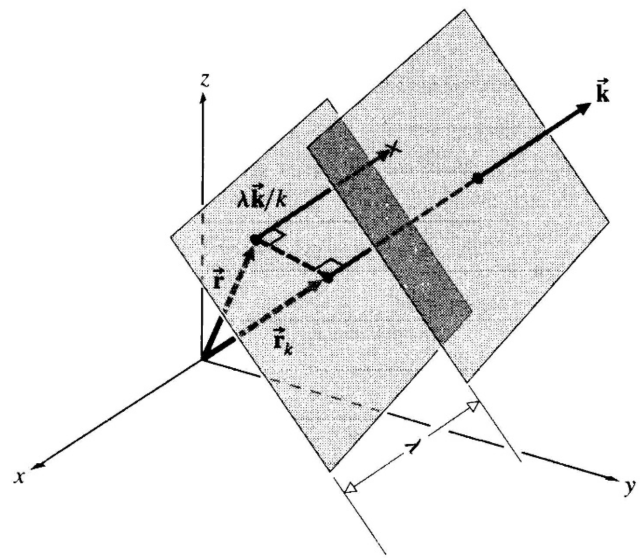
- So

$$Ae^{i\left(\vec{k}\cdot\vec{r}\mp\omega t\right)} = Ae^{i(kr_k + kdr_k \mp \omega t \mp \omega dt)} = Ae^{i(kr_k \mp \omega t)}$$

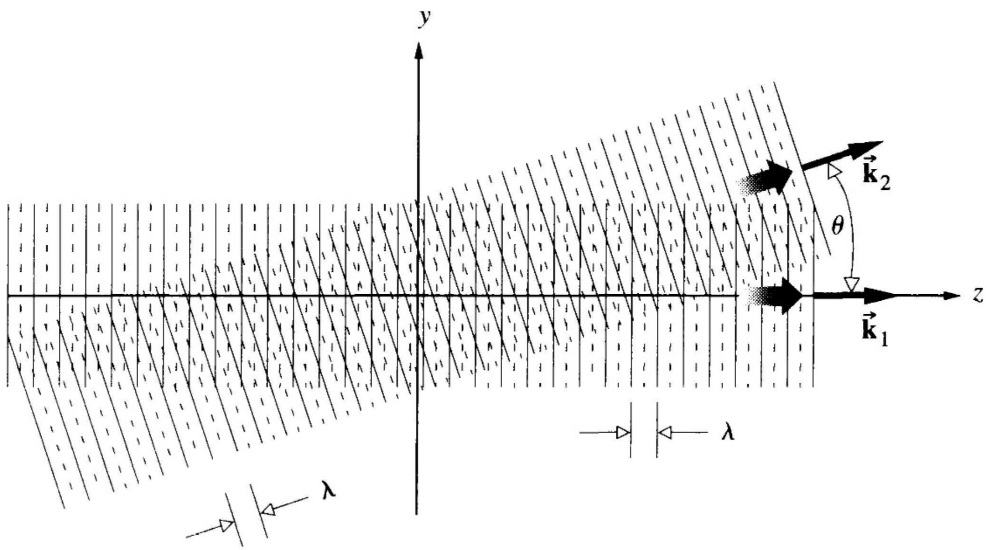
$$kdr_k = \pm\omega dt$$

- The magnitude of the wave velocity:

$$\frac{dr_k}{dt} = \pm\frac{\omega}{k} = \pm v$$



❑Example: consider the two waves shown in figure, both have the same wavelength λ.



$$\psi_1 = A_1 \cos\left(\frac{2\pi}{\lambda} z - \omega t\right)$$

$$\psi_2 = A_2 \cos\left[\frac{2\pi}{\lambda}(z\cos\theta + y\sin\theta) - \omega t\right]$$

1.7 The three-dimensional differential wave equation

- Start from $$\nabla^2\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

  $$\psi(x,y,z,t) = Ae^{i\left(\vec{k}\cdot\vec{r}\mp\omega t+\varepsilon\right)} = Ae^{i\left(k_x x+k_y y+k_z z\mp\omega t+\varepsilon\right)}$$

  Thus, $$\frac{\partial\psi}{\partial x} = ik_x\psi \Rightarrow \frac{\partial^2\psi}{\partial x^2} = -{k_x}^2\psi$$

  Similarly, $$\nabla^2\psi = -\left({k_x}^2 + {k_y}^2 + {k_z}^2\right)\psi = -k^2\psi$$

- Time derivative are given by

  $$\frac{\partial\psi}{\partial t} = -i\omega\psi \Rightarrow \frac{\partial^2\psi}{\partial t^2} = -\omega^2\psi$$

- Substitute (a) and (b) to wave equation: $$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$